

Data Analytics Capstone

**Machine Learning-Based Forecasting of Urban Air Quality Using Environmental and
Meteorological Data**

Interim Report

Vinita R. Shetty

Walsh College

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Mentor: Karthik Umesh

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GitHub Repository

GitHub Repository: [ShettyVini15/QM640-Air-Quality-Capstone: QM640 Data Analytics Capstone – Machine Learning-Based Forecasting of Urban Air Quality Using Environmental and Meteorological Data](#)

The complete project repository is available on GitHub and contains all files required to reproduce this study, including:

- **Raw Dataset:** [QM640-Air-Quality-Capstone/data/raw at main · ShettyVini15/QM640-Air-Quality-Capstone](#)
- **Preprocessed Dataset:** [QM640-Air-Quality-Capstone/data/processed at main · ShettyVini15/QM640-Air-Quality-Capstone](#)
- **Python Notebook:** [QM640-Air-Quality-Capstone/notebooks at main · ShettyVini15/QM640-Air-Quality-Capstone](#)

1. Introduction

1.1 Background

Air pollution is a major environmental and public health challenge affecting urban populations worldwide. Rapid urbanization, industrial activity, energy consumption, transportation, and other human activities contribute to elevated concentrations of harmful air pollutants in many metropolitan regions. Among these pollutants, fine particulate matter with an aerodynamic diameter of 2.5 micrometers or less (PM_{2.5}) is of particular concern because these particles can penetrate deep into the lungs and may enter the bloodstream, contributing to cardiovascular and respiratory health effects. The World Health Organization (WHO) recognizes air pollution as a major environmental health risk and provides health-based air quality guidelines for PM_{2.5} and other major pollutants (World Health Organization [WHO], 2021)

Air quality monitoring systems provide essential information about observed pollutant concentrations; however, the ability to anticipate near-future pollution levels can provide additional value for environmental monitoring and early-warning applications. Short-term forecasting of PM_{2.5} concentrations can assist environmental agencies, policymakers, and other stakeholders in understanding potential changes in air quality and planning timely responses. Machine learning provides a data-driven approach for this purpose because it can use historical pollutant, meteorological, temporal, and location-related information to model relationships associated with future PM_{2.5} concentrations.

The present study uses the Beijing Multi-Site Air Quality Dataset obtained from the UCI Machine Learning Repository. The dataset contains hourly air-quality and meteorological observations collected across 12 monitoring stations in Beijing between March 2013 and February 2017 (Chen, 2017). The original dataset consists of 420,768

observations and 18 variables, providing a substantial longitudinal dataset for developing and evaluating short-term air quality forecasting models.

Using this dataset, the study develops a one-hour-ahead PM_{2.5} forecasting framework in which information available at the current observation time is used to forecast PM_{2.5} concentration one hour later. The predictors include current-hour air pollutant concentrations, meteorological conditions, temporal variables, wind direction, and monitoring-station information. This approach enables the study to evaluate the predictive performance, temporal robustness, and interpretability of multiple machine learning regression models for short-term urban air quality forecasting.

1.2 Problem Statement

Air pollution continues to present substantial environmental and public health challenges, particularly in densely populated urban areas where elevated PM_{2.5} concentrations can adversely affect human health. Although air quality monitoring networks provide valuable measurements of current and historical pollutant concentrations, observed measurements alone do not indicate how pollution levels may change in the immediate future. Reliable short-term PM_{2.5} forecasting can therefore complement conventional monitoring by providing information about expected near-future air quality conditions and supporting environmental monitoring and early-warning applications.

The analytical challenge addressed in this study is to determine whether information available at the current observation time, including air pollutant concentrations, meteorological conditions, temporal characteristics, wind direction, and monitoring-station information, can be used to accurately forecast PM_{2.5} concentration one hour ahead. The study also addresses the need to compare multiple regression algorithms using a chronologically appropriate evaluation framework, assess whether model performance remains stable across different

temporal periods, and identify the variables that contribute most strongly to the resulting forecasts.

1.3 Purpose of the Study

The purpose of this study is to develop, evaluate, and interpret multiple machine learning regression models for one-hour-ahead PM_{2.5} forecasting using the Beijing Multi-Site Air Quality Dataset (Chen, 2017). Current-hour pollutant concentrations, meteorological conditions, temporal variables, wind direction, and monitoring-station information are used as predictor variables, while PM_{2.5}_next_hour represents the forecasting target.

Eight regression models are evaluated: Linear Regression, Decision Tree Regressor, Random Forest Regressor, XGBoost Regressor, Gradient Boosting Regressor, AdaBoost Regressor, LightGBM Regressor, and CatBoost Regressor. To preserve the temporal structure of the forecasting problem and reduce the risk of using future observations to predict earlier observations, the models are developed and evaluated using a chronological training and testing split. Predictive performance is compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

The temporal robustness of the selected model is further assessed using 5-fold expanding-window time-series cross-validation. In addition, standardized coefficient analysis and SHAP (SHapley Additive exPlanations) analysis are used to examine the relative contribution of the predictor variables to the model's forecasts. The overall purpose is therefore not only to identify a model capable of accurately forecasting PM_{2.5} concentration one hour ahead, but also to evaluate its temporal generalizability and provide an interpretable understanding of the variables contributing to short-term PM_{2.5} forecasts.

1.4 Interim Project Status (Progress Snapshot)

At the time of interim report submission, substantial progress has been made across the research, data preparation, exploratory analysis, model development, evaluation, and interpretability stages of the study. The current project status is summarized below.

Completed

- Finalized the research topic, problem statement, research objectives, and research questions.
- Conducted a comprehensive literature review and identified the research gap relevant to machine learning-based PM2.5 forecasting.
- Collected and documented the Beijing Multi-Site Air Quality Dataset from the UCI Machine Learning Repository.
- Completed data preprocessing, including missing-value treatment, duplicate assessment, outlier assessment, and feature engineering.
- Performed exploratory data analysis (EDA) to examine PM2.5 distribution, relationships among pollutant and meteorological variables, seasonal patterns, and station-wise variations in air quality.
- Created the one-hour-ahead PM2.5 forecasting target and conducted forecasting-specific exploratory analysis.
- Prepared the predictor and target variables, encoded categorical variables, and implemented a chronological training and testing split to preserve the temporal structure of the forecasting problem.
- Developed and evaluated eight machine learning regression models: Linear Regression, Decision Tree, Random Forest, XGBoost, Gradient Boosting, AdaBoost, LightGBM, and CatBoost.
- Evaluated and compared model performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

- Identified Linear Regression as the leading model overall based on the chronological testing results, with the highest testing R^2 (0.9477) and lowest testing RMSE (19.13 $\mu\text{g}/\text{m}^3$).
LightGBM achieved the lowest testing MAE (9.85 $\mu\text{g}/\text{m}^3$).
- Further assessed the temporal robustness of the selected Linear Regression model using 5-fold expanding-window time-series cross-validation.
- Conducted model interpretability analysis using standardized coefficients and SHAP (SHapley Additive exPlanations) to examine the relative contribution of predictor variables to one-hour-ahead PM2.5 forecasts.
- Prepared and maintained the GitHub repository containing the datasets, notebook, project documentation, and supporting files.
- Incorporated mentor feedback into the project analysis and Interim Report.
- Prepared the Interim Report documenting the methodology, exploratory analysis, modelling approach, results, limitations, and current findings.

In Progress

- Conducting a final consistency and quality review across the research questions, methodology, analytical results, tables, figures, and interpretation of findings.
- Ensuring that the Interim Report and GitHub repository accurately reflect the completed analysis and are appropriately documented for evaluation.

Pending

- Incorporate feedback received from the Interim Report evaluation, where applicable.
- Refine the analysis and documentation, if required, based on evaluator feedback.
- Complete the final capstone report and associated project deliverables.
- Conduct a final review of the completed capstone project for methodological consistency, reproducibility, APA 7 compliance, and presentation quality.

2. Literature Review

2.1 Machine Learning for PM2.5 Forecasting

Fine particulate matter (PM2.5) remains an important focus of air quality research because of its adverse effects on human health and its substantial spatial and temporal variability. Accurate forecasting of PM2.5 concentrations can complement conventional air quality monitoring by providing information about expected future pollution conditions and supporting early-warning and environmental decision-making applications (World Health Organization [WHO], 2021).

Machine learning has increasingly been applied to PM2.5 prediction and forecasting because air pollutant concentrations are influenced by complex relationships among historical pollution levels, meteorological conditions, temporal patterns, and location-specific characteristics. Recent comparative research has evaluated a wide range of approaches, including Linear Regression, regression trees, Support Vector Machines, ensemble methods, and neural networks. Haseeb et al. (2024), for example, compared six machine learning models for PM2.5 prediction using meteorological and chemical indicators, demonstrating the value of systematic model comparison rather than relying on a single modelling approach.

Research conducted in China has similarly demonstrated the usefulness of machine learning for PM2.5 modelling. Duan et al. (2023) compared multiple machine learning methods for predicting PM2.5 concentrations across six major Chinese urban agglomerations using meteorological information. Their study highlighted the importance of meteorological conditions and showed that model performance can vary across geographic areas and modelling approaches. More recent research has also emphasized that machine learning models may perform differently when applied to future observations rather than historical data, reinforcing the importance of appropriate temporal evaluation when the objective is forecasting

Ensemble and boosting algorithms have received particular attention in environmental prediction because they can model complex relationships among multiple predictor variables. Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost, and related methods provide alternatives to conventional linear approaches. However, greater algorithmic complexity does not automatically guarantee superior forecasting performance. Consequently, comparing simpler and more complex regression models under the same data preparation, forecasting horizon, and evaluation framework is important for identifying an appropriate model for a specific air quality forecasting problem.

2.2 Environmental and Meteorological Factors Affecting PM2.5

PM2.5 concentrations are influenced by interactions among pollutant emissions, atmospheric processes, meteorological conditions, and temporal patterns. Variables such as PM10, sulfur dioxide (SO₂), nitrogen dioxide (NO₂), carbon monoxide (CO), and ozone (O₃) can provide information about the broader pollution environment, while temperature, atmospheric pressure, dew point, rainfall, wind speed, and wind direction influence the accumulation, transport, and dispersion of particulate matter.

Recent empirical studies support the importance of combining meteorological and pollutant information when modelling PM2.5. Haseeb et al. (2024) evaluated meteorological and chemical indicators as predictors of PM2.5, while Duan et al. (2023) examined the contribution of meteorological information across multiple Chinese urban regions. Research examining PM2.5 drivers in the Beijing-Tianjin-Hebei urban agglomeration has also demonstrated the value of considering multiple environmental influences when investigating variation in particulate pollution.

Meteorological conditions are particularly important because they affect the dispersion and persistence of airborne pollutants. Wind can transport or disperse pollutants, while rainfall can contribute to atmospheric removal. Temperature, pressure, and moisture-

related variables can also be associated with atmospheric conditions that influence PM_{2.5} concentrations (Zhang et al., 2017). Recent PM_{2.5} modelling research has found that meteorological variables can provide meaningful predictive information, although their relative importance differs across locations, seasons, and modelling frameworks.

The present study uses the Beijing Multi-Site Air Quality Dataset, which provides hourly observations of six major air pollutants together with meteorological information from 12 monitoring stations between March 2013 and February 2017 (Chen, 2017). Its hourly temporal resolution and multi-station structure make it suitable for investigating short-term PM_{2.5} forecasting using pollutant, meteorological, temporal, wind-direction, and station-related predictors.

2.3 Temporal Forecasting and Model Evaluation

An important consideration in air quality forecasting is the temporal structure of the data. When the objective is to predict future pollutant concentrations, model evaluation should reflect the way the model would be used in practice. Randomly distributing observations between training and testing sets can allow observations from later periods to contribute to model training while earlier observations appear in the testing data. A chronological evaluation framework avoids this issue by training models on earlier observations and evaluating them on later, unseen observations.

This distinction is particularly relevant because strong performance on historical observations does not necessarily imply equally strong performance on future periods. Recent research comparing machine learning approaches for PM_{2.5} prediction has reported differences between performance on past and future observations, highlighting the challenges associated with temporal generalization. Short-term air quality forecasting research has also increasingly incorporated explicitly temporal and spatiotemporal modelling approaches, particularly for hourly PM_{2.5} forecasting.

For this reason, the present study uses a chronological training and testing split for one-hour-ahead PM_{2.5} forecasting. The selected model is additionally evaluated using expanding-window time-series cross-validation, in which models are repeatedly trained on earlier observations and validated on subsequent periods. This provides a more rigorous assessment of whether predictive performance remains stable across different temporal segments of the dataset.

2.4 Model Interpretability in Air Quality Forecasting

Predictive accuracy is important in environmental modelling, but understanding why a model produces its forecasts can provide additional value. Model interpretability can help identify which pollutant, meteorological, temporal, or location-related variables contribute most strongly to predictions and can also reveal whether a model relies excessively on a small number of predictors.

SHAP, or SHapley Additive exPlanations, provides a unified framework for interpreting model predictions by assigning feature-importance values to individual predictions (Lundberg & Lee, 2017). Recent PM_{2.5} research has combined machine learning with interpretability methods to examine the variables driving model predictions. For example, recent work using feature engineering and SHAP demonstrated how interpretability analysis can complement predictive performance by identifying the variables contributing most strongly to PM_{2.5} forecasts. Other recent air quality research has similarly used SHAP to provide transparent assessments of how meteorological and co-pollutant variables influence model predictions.

In the present study, interpretability is examined using standardized coefficients and SHAP analysis for the selected Linear Regression model. Standardized coefficients allow predictors measured on different scales to be compared more meaningfully, while SHAP provides an additional assessment of feature contributions across observations. Together,

these approaches support both predictive evaluation and interpretation of the factors contributing to one-hour-ahead PM_{2.5} forecasts.

2.5 Research Gap

Existing research demonstrates substantial progress in the application of machine learning to PM_{2.5} prediction and forecasting. Studies have compared different algorithms, incorporated pollutant and meteorological variables, investigated spatial and temporal relationships, and increasingly applied explainability techniques. Recent comparative research nevertheless indicates that model performance depends on the predictors, location, forecasting design, and evaluation strategy used, and that more complex algorithms do not necessarily provide the best performance in every setting.

Furthermore, while previous research has demonstrated the effectiveness of ensemble learning methods, there remains a need for systematic comparison of traditional regression approaches and advanced ensemble algorithms using standardized evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Such comparisons can provide valuable insights into model selection for environmental forecasting.

Three areas are particularly relevant to the present study. First, meaningful comparison requires multiple regression algorithms to be evaluated using the same dataset, forecasting horizon, predictor set, and performance metrics. Second, a forecasting study should preserve the temporal ordering of observations so that models are evaluated on genuinely later periods rather than relying solely on random train-test partitions. Third, predictive performance should be complemented by interpretability analysis so that the contribution of pollutant, meteorological, temporal, and location-related variables can be examined.

To address these considerations, the present study develops and compares eight regression models: Linear Regression, Decision Tree, Random Forest, XGBoost, Gradient Boosting, AdaBoost, LightGBM, and CatBoost, using a common one-hour-ahead PM2.5 forecasting framework. The models are evaluated using MAE, RMSE, and R^2 on a chronological testing period. The leading model is further examined using 5-fold expanding-window time-series cross-validation, standardized coefficient analysis, and SHAP. This integrated approach enables the study to compare predictive performance, temporal robustness, and model interpretability within a single forecasting framework.

3. Scope, Research Objectives and Research Questions

3.1 Scope of the Study

The scope of this study is limited to one-hour-ahead forecasting of PM2.5 concentration using the Beijing Multi-Site Air Quality Dataset. The analysis uses hourly air quality and meteorological observations collected from 12 monitoring stations in Beijing, China, between March 2013 and February 2017. Current-hour pollutant concentrations, meteorological conditions, temporal variables, wind direction, and monitoring-station information are considered as predictors of PM2.5 concentration one hour ahead.

The study develops and compares eight supervised machine learning regression models: Linear Regression, Decision Tree Regressor, Random Forest Regressor, XGBoost Regressor, Gradient Boosting Regressor, AdaBoost Regressor, LightGBM Regressor, and CatBoost Regressor. Model performance is evaluated using MAE, RMSE, and R^2 within a chronological evaluation framework. The temporal robustness of the selected model is further assessed using expanding-window time-series cross-validation, while standardized coefficients and SHAP analysis are used to interpret predictor contributions.

The study is restricted to the variables and historical period available in the selected dataset and does not incorporate external factors such as traffic volume, industrial activity,

real-time emissions, satellite observations, or data from locations outside Beijing. The study focuses on predictive forecasting rather than establishing causal relationships between environmental variables and PM_{2.5} concentration.

3.2 Research Objectives

The primary objective of this study is to develop an accurate and reliable machine learning model for forecasting PM_{2.5} concentration one hour ahead using pollutant, meteorological, temporal, wind-direction, and monitoring-station information.

The specific objectives are:

1. To identify the pollutant and meteorological variables that influence one-hour-ahead PM_{2.5} concentration.
2. To analyze the relationships between current-hour pollutant and meteorological variables and one-hour-ahead PM_{2.5} concentration using exploratory data analysis and statistical techniques.
3. To develop and compare multiple machine learning regression models for forecasting PM_{2.5} concentration one hour ahead.
4. To evaluate the predictive performance of the developed models using MAE, RMSE, and R^2 within a chronological evaluation framework.
5. To assess the temporal robustness of the selected model using expanding-window time-series cross-validation.
6. To identify the variables contributing most strongly to the selected model using standardized coefficients and SHAP analysis.
7. To recommend the most suitable machine learning model for supporting short-term air quality forecasting and environmental decision-making.

3.3 Research Questions

The study seeks to answer the following research questions:

- **RQ1.** Which current-hour pollutant and meteorological variables are most strongly associated with PM2.5 concentration one hour ahead?
- **RQ2.** How accurately can machine learning regression models forecast PM2.5 concentration one hour ahead using current-hour pollutant, meteorological, temporal, wind-direction, and monitoring-station information?
- **RQ3.** Which machine learning regression model provides the best predictive performance for one-hour-ahead PM2.5 forecasting?
- **RQ4.** Which predictor variables contribute most strongly to the forecasts of the selected machine learning model?

3.3 Null and Alternative Hypotheses

RQ1. Which current-hour pollutant and meteorological variables are most strongly associated with PM2.5 concentration one hour ahead?

- **H₀₁:** There is no meaningful association between current-hour pollutant and meteorological variables and one-hour-ahead PM2.5 concentration.
- **H₁₁:** At least one current-hour pollutant or meteorological variable has a meaningful association with one-hour-ahead PM2.5 concentration.

RQ2. How accurately can machine learning regression models forecast PM2.5 concentration one hour ahead using current-hour pollutant, meteorological, temporal, wind-direction, and monitoring-station information?

- **H₀₂:** Machine learning regression models do not provide meaningful predictive capability for forecasting PM2.5 concentration one hour ahead using the selected predictor variables.
- **H₁₂:** Machine learning regression models provide meaningful predictive capability for forecasting PM2.5 concentration one hour ahead using the selected predictor variables.

RQ3. Which machine learning regression model provides the best predictive performance for one-hour-ahead PM2.5 forecasting?

- **H₀₃:** There is no meaningful difference in predictive performance among the evaluated machine learning regression models.
- **H₁₃:** At least one evaluated machine learning regression model demonstrates better predictive performance than the others.

RQ4. Which predictor variables contribute most strongly to the forecasts of the selected machine learning model?

- **H₀₄:** None of the predictor variables makes a meaningful contribution to the forecasts produced by the selected machine learning model.
- **H₁₄:** At least one predictor variable makes a meaningful contribution to the forecasts produced by the selected machine learning model.

4. Sample Size Calculation

The minimum sample size for this study was determined using the Confidence Interval (CI) method. A 95% confidence level was selected because it is a commonly accepted standard in academic research, corresponding to a Z-value of 1.96. An estimated population proportion of 0.50 was used because the true population proportion was unknown; this value provides a conservative estimate by producing the maximum required sample size. A 5% margin of error was selected to provide a reasonable level of precision for the analysis. The minimum sample size was calculated using the following formula:

$$n = \frac{Z^2 \times p(1 - p)}{e^2}$$

$$n = \frac{(1.96)^2 \times 0.50(1 - 0.50)}{(0.05)^2}$$

$$n = 384.16 \approx 385 \text{ observations}$$

Therefore, the minimum required sample size was 385 observations. Since all four research questions are investigated using the same underlying dataset, the same minimum sample size requirement applies to RQ1 through RQ4.

Following preprocessing and construction of the one-hour-ahead PM2.5 forecasting target, 420,734 aligned observations were available for forecasting analysis, substantially exceeding the minimum requirement. A research-question-wise summary of the minimum sample size is presented in Appendix F (Table F1).

4.1 Minimum Sample Size Computation per Research Question

The calculated minimum sample size of 385 observations applies separately to RQ1, RQ2, RQ3, and RQ4 because all four research questions are addressed using the same underlying dataset and forecasting analysis. The available forecasting dataset of 420,734 observations substantially exceeds this requirement for each research question.

5. Data Description

5.1 Data Source

The study uses the Beijing Multi-Site Air Quality Dataset (Chen, 2017), obtained from the UCI Machine Learning Repository. The dataset contains hourly air quality and meteorological observations collected from 12 monitoring stations in Beijing, China, between 1 March 2013 and 28 February 2017. The original dataset comprises 420,768 observations and 18 variables, including pollutant concentrations, meteorological measurements, temporal variables, wind direction, and monitoring-station identifiers. The dataset was selected because its hourly temporal resolution, multi-station coverage, and combination of pollutant and meteorological variables make it suitable for developing and evaluating models for one-hour-ahead PM2.5 forecasting.

5.2 Dataset Description

Each observation represents air quality and meteorological conditions recorded at a specific monitoring station during a particular hour. The dataset includes temporal variables (year, month, day, and hour); pollutant measurements (PM2.5, PM10, SO₂, NO₂, CO, and O₃); meteorological variables including temperature (TEMP), atmospheric pressure (PRES), dew point (DEWP), rainfall (RAIN), wind direction (wd), and wind speed (WSPM); and the monitoring-station identifier.

Initial data exploration examined the dataset structure, variable types, descriptive statistics, duplicate records, and missing values. Missing values were identified in several pollutant and meteorological variables and were subsequently addressed during data preprocessing. For the forecasting analysis, a target variable, PM2.5_next_hour, was created to represent PM2.5 concentration one hour ahead. Following target construction and alignment of consecutive hourly observations within each monitoring station, 420,734 observations were available for the forecasting analysis.

Supporting information on the dataset, including sample records, descriptive statistics, variable structure, and missing-value assessment, is presented in Appendix A.

5.3 Data Availability and Ethical Considerations

The Beijing Multi-Site Air Quality Dataset is publicly available through the UCI Machine Learning Repository and contains environmental monitoring data without personally identifiable information. Therefore, the study presents minimal ethical concerns related to individual privacy and confidentiality. The dataset was used solely for academic research, with appropriate acknowledgment of the original source. The use of a publicly accessible and documented dataset also supports transparency and reproducibility of the analysis.

6. Data Preprocessing

Data preprocessing was performed to prepare the Beijing Multi-Site Air Quality Dataset for one-hour-ahead PM_{2.5} forecasting. The integrated dataset was systematically examined for missing values, duplicate records, potential outliers, and data inconsistencies. Preprocessing procedures were designed to preserve the chronological structure of the data and avoid using future information when preparing earlier observations.

6.1 Data Quality Assessment

The integrated dataset was initially assessed for completeness, consistency, and suitability for machine learning analysis. The assessment included dataset dimensions, variable names, data types, descriptive statistics, missing values, and duplicate records. The original integrated dataset contained 420,768 hourly observations and 18 variables, including temporal variables, air pollutant measurements, meteorological variables, wind direction, and monitoring-station identifiers.

The assessment confirmed that the dataset was appropriately structured for the forecasting analysis. However, several pollutant and meteorological variables contained missing observations that required treatment before exploratory analysis and model development. Supporting outputs from the data quality assessment are presented in Appendix A.

6.2 Missing Value Analysis

Missing values were identified in the pollutant variables PM_{2.5}, PM₁₀, SO₂, NO₂, CO, and O₃ and in the meteorological variables TEMP, PRES, DEWP, RAIN, and WSPM. Missing observations were also present in the categorical wind-direction variable (wd).

Among the pollutant variables, CO contained the highest number of missing observations, followed by O₃ and NO₂. The proportion of missing values remained below 5% for each variable, indicating that most of the original observations were complete.

6.3 Missing Value Treatment

To preserve the temporal structure of the forecasting problem, observations were first sorted chronologically within each monitoring station. Missing numerical values were then treated using station-wise forward filling, whereby a missing observation was replaced using the most recently available preceding value from the same monitoring station. Missing wind-direction values were treated using the same station-wise forward-filling approach.

This method was selected because it uses previously observed information rather than future observations, making it consistent with the one-hour-ahead forecasting framework. After forward filling, 22 observations still contained missing values because no previous observation was available within the corresponding station. These observations were removed.

Following missing-value treatment, all required variables contained zero missing values, and the dataset contained 420,746 observations.

6.4 Duplicate Record Analysis

Duplicate records were examined to ensure that each row represented a unique observation. The analysis identified no duplicate records in the integrated dataset. Therefore, no observations were removed during duplicate assessment.

6.5 Outlier Detection and Treatment

Potential outliers in the numerical variables were assessed using the Interquartile Range (IQR) method. Extreme observations were identified in several pollutant and meteorological variables, particularly SO₂, CO, WSPM, PM_{2.5}, O₃, RAIN, PM₁₀, and NO₂.

The identified potential outliers were retained rather than automatically removed because extreme pollutant concentrations and meteorological conditions may represent genuine environmental events rather than data errors. Retaining these observations allowed the forecasting models to learn from a broader range of real-world air quality conditions.

6.6 Feature Engineering

A datetime variable was created by combining the year, month, day, and hour variables into a single timestamp. The monitoring-station identifier was retained to preserve location-related information. The datetime feature provided a consistent temporal reference for arranging observations chronologically and supported the subsequent construction of the one-hour-ahead forecasting target, chronological train-test split, and time-series cross-validation.

For the forecasting analysis, a new target variable, `PM2.5_next_hour`, was created to represent PM2.5 concentration one hour ahead. Target values were aligned within each monitoring station so that current-hour predictor information corresponded to the subsequent hourly PM2.5 observation. Only valid consecutive hourly observations within the same station were retained, resulting in 420,734 aligned forecasting observations.

6.7 Categorical Variable Encoding

The categorical variables representing monitoring station and wind direction were converted into numerical dummy variables for machine learning model development. One category from each categorical variable was omitted as the reference category to avoid redundant dummy variables and perfect multicollinearity.

Following categorical encoding, the forecasting dataset contained 420,734 observations and 41 predictor variables. The encoding procedure changed only the representation of the categorical variables and did not remove additional observations.

6.8 Final Preprocessed Dataset

Following missing-value treatment and datetime feature creation, the preprocessed dataset contained 420,746 observations and 19 variables, with zero remaining missing values and no duplicate records. After construction and alignment of the one-hour-ahead PM2.5 forecasting target, 420,734 observations were available for forecasting analysis.

The final predictor dataset contained 41 features after categorical encoding and was subsequently used for chronological training and testing, model development, and evaluation. Detailed preprocessing outputs, including missing-value analysis, outlier assessment, feature engineering, target construction, and final dataset verification, are presented in Appendix B.

7. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine the distribution, temporal patterns, spatial variation, and relationships among the air pollutant and meteorological variables before model development. The analysis included PM2.5 distribution, correlation analysis, monthly variation, station-wise variation, meteorological relationships, and forecasting-specific analysis.

The PM2.5 distribution was strongly right-skewed, with a skewness of 2.01, indicating that most observations occurred at lower to moderate concentrations while fewer observations represented high-pollution events. These extreme observations were retained because they may represent genuine pollution episodes relevant to one-hour-ahead forecasting.

Correlation analysis showed that current-hour PM2.5 had strong positive correlations with PM10 ($r = 0.88$) and CO ($r = 0.77$), followed by NO₂ ($r = 0.65$) and SO₂ ($r = 0.47$). Among the meteorological variables, wind speed showed the strongest negative relationship with PM2.5 ($r = -0.27$), while temperature showed a weaker negative relationship ($r = -0.13$). These findings indicate that PM2.5 is more strongly associated with other pollutant concentrations than with individual meteorological variables.

Monthly analysis revealed a clear seasonal pattern. PM2.5 concentrations were generally higher during the winter period and lower during the summer, with the highest monthly average occurring in December and the lowest in August. This pattern supported the inclusion of temporal information in the forecasting models.

Forecasting-specific exploratory analysis was also conducted to examine relationships between current-hour predictors and the one-hour-ahead PM2.5 target. The analysis showed a particularly strong relationship between current-hour PM2.5 and next-hour PM2.5, supporting the inclusion of current PM2.5 as a key forecasting predictor. Detailed EDA tables and visualizations are presented in Appendix C.

8. Model Development

Following data preprocessing and exploratory data analysis, the forecasting dataset was prepared for supervised machine learning. The modelling framework was designed specifically for one-hour-ahead PM2.5 forecasting, with particular attention to predictor selection, temporal integrity, comparative model evaluation, and interpretability.

8.1 Target and Predictor Features

The target variable was defined as PM2.5 concentration one hour ahead (PM2.5_next_hour). The predictor variables represented information available at the current hour and included air pollutant measurements, meteorological conditions, temporal information, wind direction, and monitoring-station location.

The predictor set included current-hour PM2.5, PM10, SO₂, NO₂, CO, O₃, TEMP, PRES, DEWP, RAIN, WSPM, month, and hour, together with encoded wind-direction and monitoring-station variables. Categorical variables representing wind direction and monitoring station were converted into dummy variables. Following categorical encoding, the final modelling dataset contained 41 predictor features.

Current-hour PM2.5 was retained as a predictor because the study forecasts PM2.5 at the subsequent hour rather than estimating the concentration at the same hour. Therefore, current-hour PM2.5 represents information available at the time a one-hour-ahead forecast would be generated. A detailed list of the predictor features is provided in Appendix D.

8.2 Choice of Models and Justification

Eight regression algorithms were developed to compare models with different levels of complexity and modelling assumptions.

Linear Regression was included as an interpretable baseline model for assessing whether the forecasting relationship could be represented effectively using a relatively simple linear framework. Decision Tree Regression was selected because it can capture nonlinear relationships and interactions without requiring linearity assumptions.

Random Forest Regression was included as an ensemble tree-based method capable of reducing the instability and overfitting associated with individual decision trees (Breiman, 2001). Gradient Boosting Regression was selected because it sequentially improves predictions by correcting errors from preceding learners and can capture complex nonlinear relationships (Friedman, 2001). XGBoost Regression was included as an efficient and scalable implementation of gradient-boosted trees (Chen & Guestrin, 2016).

AdaBoost Regression was included to evaluate an adaptive boosting approach that places greater emphasis on observations that are more difficult to predict. LightGBM Regression was selected because of its computational efficiency and suitability for large structured datasets. CatBoost Regression was included as an additional gradient-boosting approach to provide a broader comparison of advanced ensemble methods.

Together, these eight models enabled comparison between an interpretable linear baseline, an individual nonlinear tree model, and multiple ensemble and boosting approaches for one-hour-ahead PM_{2.5} forecasting.

8.3 Chronological Training and Testing Strategy

Because the study focuses on forecasting future PM_{2.5} concentrations, a chronological train-test split was used rather than random sampling. Earlier observations were assigned to the training dataset and later observations to the testing dataset. This ensured that

models were trained using historical information and evaluated on subsequent unseen observations, providing a more realistic assessment of one-hour-ahead forecasting performance.

The selected model was further evaluated using five-fold expanding-window time-series cross-validation. Each validation period occurred after its corresponding training period, preserving temporal integrity and allowing the stability of model performance to be examined across different time periods. This approach is consistent with the one-hour-ahead forecasting objective.

8.4 Model Evaluation Metrics and Practical Relevance

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) (Pedregosa et al., 2011).

MAE represents the average magnitude of forecasting errors and provides an intuitive indication of how far predicted next-hour PM_{2.5} concentrations deviate from observed concentrations. From an operational perspective, lower MAE indicates greater average forecasting accuracy, which can improve the usefulness of forecasts for routine air quality monitoring.

RMSE gives greater weight to larger forecasting errors. This is particularly relevant for air quality applications because substantial forecasting errors during elevated pollution periods may reduce the effectiveness of early-warning, public-health advisory, and pollution-management decisions.

R^2 represents the proportion of variation in one-hour-ahead PM_{2.5} concentration explained by the model and provides an overall measure of predictive strength. A higher R^2 indicates that the forecasting model captures a greater proportion of temporal variation in future PM_{2.5} levels.

Together, the three metrics provide complementary measures of forecasting performance and support both statistical model comparison and assessment of the practical usefulness of the models for environmental monitoring and evidence-based decision-making.

9. Preliminary Results

The comparative evaluation identified Linear Regression as the leading model overall on the chronological testing dataset. The model achieved a Test MAE of 10.16 $\mu\text{g}/\text{m}^3$, Test RMSE of 19.13 $\mu\text{g}/\text{m}^3$, and Test R^2 of 0.9477. Although LightGBM achieved the lowest Test MAE of 9.85 $\mu\text{g}/\text{m}^3$, Linear Regression achieved the highest Test R^2 and lowest Test RMSE and was therefore selected for further robustness and interpretability analysis.

Five-fold expanding-window time-series cross-validation further supported the stability of the selected model. Linear Regression achieved a mean MAE of 11.93 $\mu\text{g}/\text{m}^3$, mean RMSE of 20.43 $\mu\text{g}/\text{m}^3$, and mean R^2 of 0.9320. R^2 values ranged from 0.9126 to 0.9468, with a standard deviation of 0.0138, indicating relatively consistent predictive performance across different temporal periods.

Standardized coefficient and SHAP analyses identified current-hour PM_{2.5} as the dominant predictor of PM_{2.5} concentration one hour ahead. The mean absolute SHAP value for current-hour PM_{2.5} was 57.87, substantially exceeding the contributions of the remaining predictors. TEMP, DEWP, PM₁₀, NO₂, hour, and WSPM provided additional predictive information.

Detailed model comparison results, cross-validation results, standardized coefficient analysis, and SHAP visualisations are presented in Appendix E.

From a practical decision-making perspective, accurate one-hour-ahead PM_{2.5} forecasts can support environmental monitoring agencies and public health authorities by providing earlier indication of changing pollution conditions. Lower forecasting errors can improve the reliability of short-term air quality alerts, while identifying influential predictors

can help stakeholders understand the environmental conditions associated with near-term changes in PM_{2.5} concentration. The forecasting framework therefore provides both predictive and interpretive value for short-term air quality monitoring and pollution management decisions.

10. Conclusion and Future Work

This interim study developed and evaluated eight machine learning regression models for one-hour-ahead PM_{2.5} forecasting using the Beijing Multi-Site Air Quality Dataset. The chronological evaluation identified Linear Regression as the leading model overall, achieving a Test R^2 of 0.9477 and Test RMSE of 19.13 $\mu\text{g}/\text{m}^3$. Time-series cross-validation further demonstrated consistent performance across different temporal periods.

Model interpretation showed that current-hour PM_{2.5} was the dominant predictor of PM_{2.5} concentration one hour ahead, while pollutant, meteorological, temporal, and location-related variables provided additional predictive information. These findings demonstrate the potential of an interpretable machine learning framework for short-term air quality forecasting.

The remaining work will focus on final report refinement and incorporation of evaluator feedback. Future research may evaluate the forecasting framework using data from other geographical regions, additional environmental and traffic-related variables, real-time data streams, and advanced time-series or deep learning approaches.

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Appendix A

Dataset Description and Initial Data Assessment

Table A1

Data Dictionary for the Beijing Multi-Site Air Quality Dataset

Variable	Description	Data Type	Role in Study
No	Record identifier	Numeric	Identifier, excluded from modelling
year	Year of observation	Numeric	Temporal variable
month	Month of observation	Numeric	Temporal predictor
day	Day of month	Numeric	Temporal variable
hour	Hour of day	Numeric	Temporal predictor
PM2.5	Current-hour fine particulate matter concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
PM10	Current-hour particulate matter $\leq 10 \mu\text{m}$ concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
SO ₂	Current-hour sulfur dioxide concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
NO ₂	Current-hour nitrogen dioxide concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
CO	Current-hour carbon monoxide concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
O ₃	Current-hour ozone concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
TEMP	Temperature ($^{\circ}\text{C}$)	Numeric	Meteorological predictor
PRES	Atmospheric pressure (hPa)	Numeric	Meteorological predictor
DEWP	Dew point temperature ($^{\circ}\text{C}$)	Numeric	Meteorological predictor
RAIN	Precipitation (mm)	Numeric	Meteorological predictor
wd	Wind direction	Categorical	Meteorological predictor
WSPM	Wind speed (m/s)	Numeric	Meteorological predictor
station	Monitoring-station name	Categorical	Location-related predictor

Note. Variable names and descriptions are based on the Beijing Multi-Site Air Quality Dataset (Chen, 2017). For the forecasting analysis, an engineered target variable, PM2.5_next_hour, was created to represent PM2.5 concentration one hour ahead. Current-hour PM2.5 was therefore used as a predictor rather than as the forecasting target. The record identifier (No) and variables not included in the final predictor set were excluded before model development

... First Five Records

	No	year	month	day	hour	PM2.5	PM10	SO2	NO2	CO	O3	TEMP	PRES	DEWP	RAIN	wd	WSPM	station
0	1	2013	3	1	0	4.0	4.0	4.0	7.0	300.0	77.0	-0.7	1023.0	-18.8	0.0	NNW	4.4	Aotizhongxin
1	2	2013	3	1	1	8.0	8.0	4.0	7.0	300.0	77.0	-1.1	1023.2	-18.2	0.0	N	4.7	Aotizhongxin
2	3	2013	3	1	2	7.0	7.0	5.0	10.0	300.0	73.0	-1.1	1023.5	-18.2	0.0	NNW	5.6	Aotizhongxin
3	4	2013	3	1	3	6.0	6.0	11.0	11.0	300.0	72.0	-1.4	1024.5	-19.4	0.0	NW	3.1	Aotizhongxin
4	5	2013	3	1	4	3.0	3.0	12.0	12.0	300.0	72.0	-2.0	1025.2	-19.5	0.0	N	2.0	Aotizhongxin

Figure A1

First Five Records of the Beijing Multi-Site Air Quality Dataset

Note. The figure displays the first five observations of the Beijing Multi-Site Air Quality Dataset, illustrating the dataset structure, variable names, and sample values used in this study.

df.shape

(420768, 18)

Figure A2

Dataset Dimensions of the Beijing Multi-Site Air Quality Dataset

```

*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 420768 entries, 0 to 420767
Data columns (total 18 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   No           420768 non-null  int64
1   year         420768 non-null  int64
2   month        420768 non-null  int64
3   day          420768 non-null  int64
4   hour         420768 non-null  int64
5   PM2.5        412029 non-null  float64
6   PM10         414319 non-null  float64
7   SO2          411747 non-null  float64
8   NO2          408652 non-null  float64
9   CO           400067 non-null  float64
10  O3           407491 non-null  float64
11  TEMP         420370 non-null  float64
12  PRES         420375 non-null  float64
13  DEWP         420365 non-null  float64
14  RAIN         420378 non-null  float64
15  wd           418946 non-null  object
16  WSPM         420450 non-null  float64
17  station      420768 non-null  object
dtypes: float64(11), int64(5), object(2)
memory usage: 57.8+ MB

```

Figure A3

Dataset Information Showing Variable Names, Data Types, and Non-Null Values

	count	mean	std	min	25%	50%	75%	max
No	420768.0	17532.500000	10122.116943	1.0000	8766.75	17532.5	26298.25	35064.0
year	420768.0	2014.662560	1.177198	2013.0000	2014.00	2015.0	2016.00	2017.0
month	420768.0	6.522930	3.448707	1.0000	4.00	7.0	10.00	12.0
day	420768.0	15.729637	8.800102	1.0000	8.00	16.0	23.00	31.0
hour	420768.0	11.500000	6.922195	0.0000	5.75	11.5	17.25	23.0
PM2.5	412029.0	79.793428	80.822391	2.0000	20.00	55.0	111.00	999.0
PM10	414319.0	104.602618	91.772426	2.0000	36.00	82.0	145.00	999.0
SO2	411747.0	15.830835	21.650603	0.2856	3.00	7.0	20.00	500.0
NO2	408652.0	50.638586	35.127912	1.0265	23.00	43.0	71.00	290.0
CO	400067.0	1230.766454	1160.182716	100.0000	500.00	900.0	1500.00	10000.0
O3	407491.0	57.372271	56.661607	0.2142	11.00	45.0	82.00	1071.0
TEMP	420370.0	13.538976	11.436139	-19.9000	3.10	14.5	23.30	41.6
PRES	420375.0	1010.746982	10.474055	982.4000	1002.30	1010.4	1019.00	1042.8
DEWP	420365.0	2.490822	13.793847	-43.4000	-8.90	3.1	15.10	29.1
RAIN	420378.0	0.064476	0.821004	0.0000	0.00	0.0	0.00	72.5
WSPM	420450.0	1.729711	1.246386	0.0000	0.90	1.4	2.20	13.2

Figure A4

Summary Statistics of Numerical Variables

No	0
year	0
month	0
day	0
hour	0
PM2.5	8739
PM10	6449
SO2	9021
NO2	12116
CO	20701
O3	13277
TEMP	398
PRES	393
DEWP	403
RAIN	390
wd	1822
WSPM	318
station	0
dtype: int64	

Figure A5

Missing Values Across Dataset Variables

Appendix B

Data Preprocessing and Feature Engineering

Table B1

Missing Value Analysis Before Preprocessing

	Missing Values	Percentage (%)
CO	20701	4.919813
O3	13277	3.155421
NO2	12116	2.879497
SO2	9021	2.143937
PM2.5	8739	2.076916
PM10	6449	1.532674
wd	1822	0.433018
DEWP	403	0.095777
TEMP	398	0.094589
PRES	393	0.093401
RAIN	390	0.092688
WSPM	318	0.075576

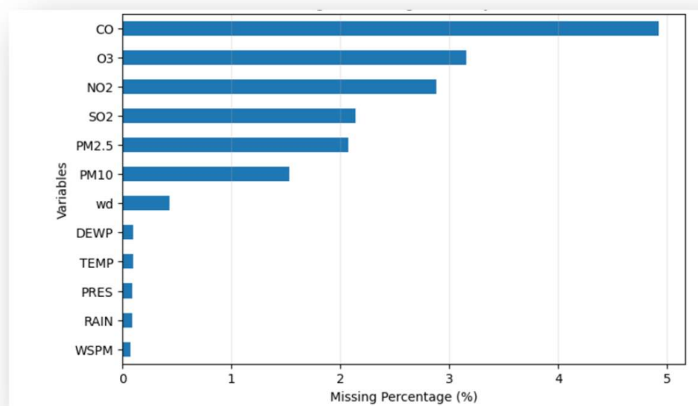


Figure B1

Percentage of Missing Values by Variable

Number of duplicate records: 0

Figure B2

Duplicate Record Verification

Table B2

Missing-Value Verification After Treatment

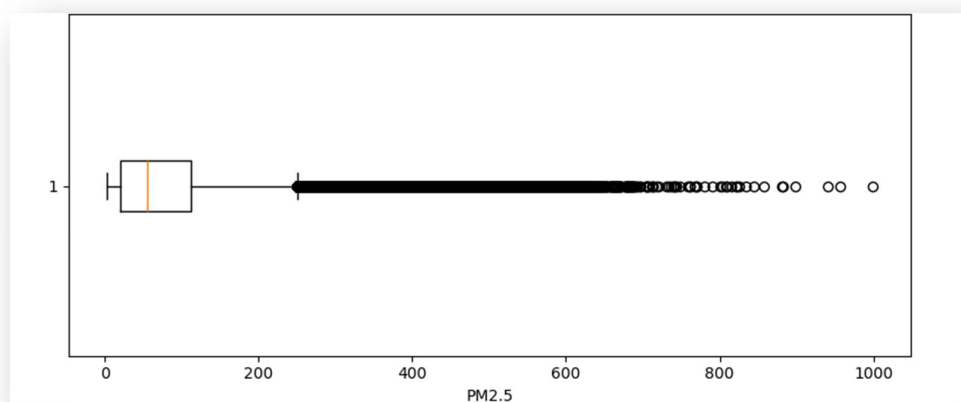
Missing Values	
No	0
year	0
month	0
day	0
hour	0
PM2.5	0
PM10	0
SO2	0
NO2	0
CO	0
O3	0
TEMP	0
PRES	0
DEWP	0
RAIN	0
wd	0
WSPM	0
station	0

Table B3

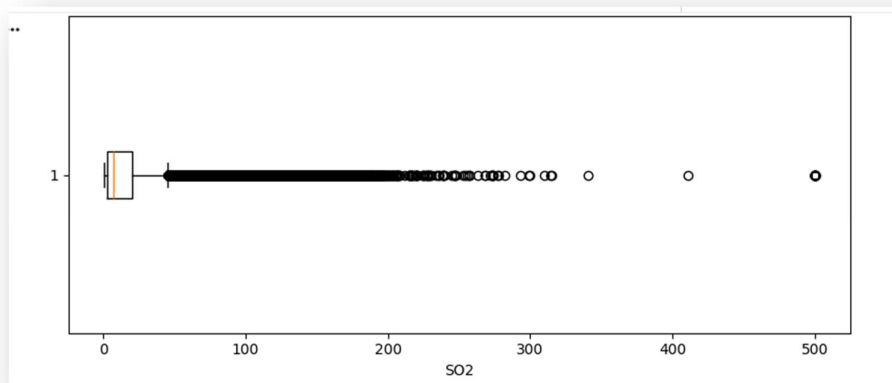
Outlier Summary Using the Interquartile Range (IQR) Method

	Variable	Number of Potential Outliers	Percentage (%)
0	PM2.5	19168	4.56
1	PM10	14777	3.51
2	SO2	36572	8.69
3	NO2	7311	1.74
4	CO	30279	7.20
5	O3	17308	4.11
6	TEMP	0	0.00
7	PRES	0	0.00
8	DEWP	0	0.00
9	RAIN	16525	3.93
10	WSPM	23111	5.49

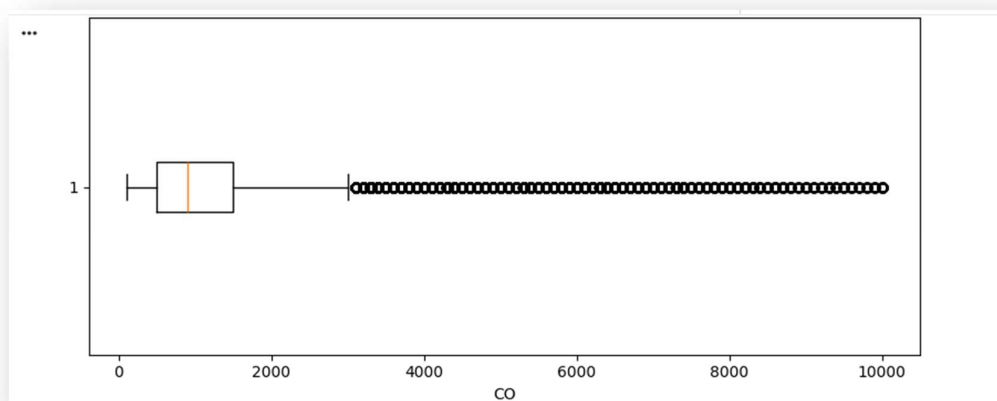
Note. Selected boxplots are presented for variables exhibiting notable potential outlier patterns based on the IQR assessment. The complete outlier assessment for all numerical variables is provided in Table B3.

**Figure B3**

Boxplot of PM2.5 Concentration

**Figure B4**

Boxplot of SO₂ Concentration

**Figure B5**

Boxplot of CO Concentration

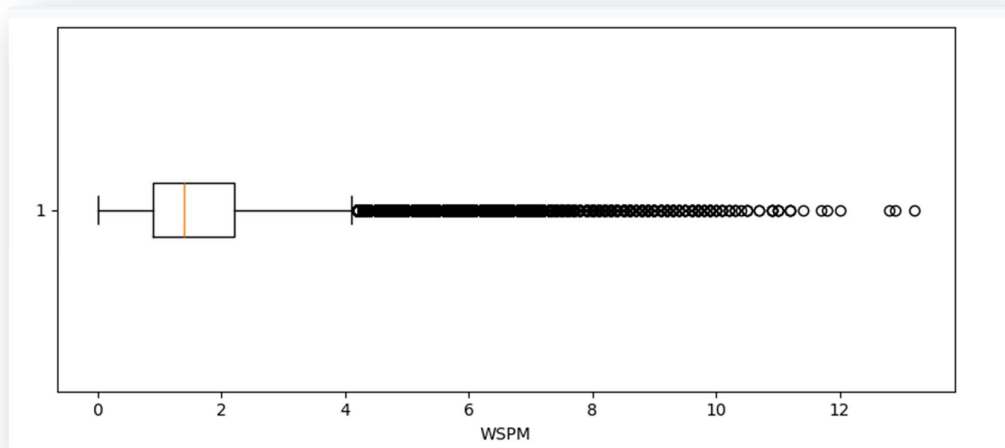


Figure B6

Boxplot of Wind Speed (WSPM)

	datetime	station
0	2013-03-01 00:00:00	Aotizhongxin
1	2013-03-01 01:00:00	Aotizhongxin
2	2013-03-01 02:00:00	Aotizhongxin
3	2013-03-01 03:00:00	Aotizhongxin
4	2013-03-01 04:00:00	Aotizhongxin
Earliest Date and Time: 2013-03-01 00:00:00		
Latest Date and Time: 2017-02-28 23:00:00		

Figure B7

Feature Engineering: Creation of the Datetime Variable

Dataset Shape: (420746, 19)

Missing Values After Preprocessing:

Missing Values

No	0
year	0
month	0
day	0
hour	0
PM2.5	0
PM10	0
SO2	0
NO2	0
CO	0
O3	0
TEMP	0
PRES	0
DEWP	0
RAIN	0
wd	0
WSPM	0
station	0
datetime	0

Total Remaining Missing Values: 0

First Five Rows of the Preprocessed Dataset:

	No	year	month	day	hour	PM2.5	PM10	SO2	NO2	CO	O3	TEMP	PRES	DEWP	RAIN	wd	WSPM	station	datetime
0	1	2013	3	1	0	4.0	4.0	4.0	7.0	300.0	77.0	-0.7	1023.0	-18.8	0.0	NNW	4.4	Aotizhongxin	2013-03-01 00:00:00
1	2	2013	3	1	1	8.0	8.0	4.0	7.0	300.0	77.0	-1.1	1023.2	-18.2	0.0	N	4.7	Aotizhongxin	2013-03-01 01:00:00
2	3	2013	3	1	2	7.0	7.0	5.0	10.0	300.0	73.0	-1.1	1023.5	-18.2	0.0	NNW	5.6	Aotizhongxin	2013-03-01 02:00:00
3	4	2013	3	1	3	6.0	6.0	11.0	11.0	300.0	72.0	-1.4	1024.5	-19.4	0.0	NW	3.1	Aotizhongxin	2013-03-01 03:00:00
4	5	2013	3	1	4	3.0	3.0	12.0	12.0	300.0	72.0	-2.0	1025.2	-19.5	0.0	N	2.0	Aotizhongxin	2013-03-01 04:00:00

Figure B8

Final Preprocessed Beijing Multi-Site Air Quality Dataset

Appendix C

Exploratory Data Analysis

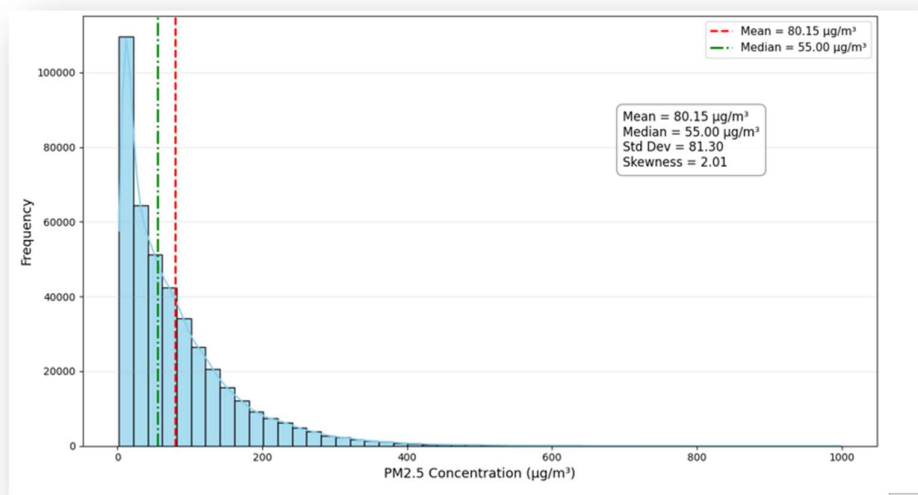


Figure C1

Distribution of PM2.5 Concentration

Table C1

Correlation Matrix of Air Pollutants and Meteorological Variables

	PM2.5	PM10	SO2	NO2	CO	O3	TEMP	PRES	DEWP	RAIN	WSPM
PM2.5	1.00	0.88	0.47	0.65	0.77	-0.14	-0.13	0.02	0.11	-0.01	-0.27
PM10	0.88	1.00	0.45	0.64	0.69	-0.11	-0.10	-0.02	0.07	-0.03	-0.18
SO2	0.47	0.45	1.00	0.48	0.51	-0.16	-0.32	0.22	-0.26	-0.04	-0.11
NO2	0.65	0.64	0.48	1.00	0.68	-0.45	-0.27	0.17	-0.03	-0.04	-0.39
CO	0.77	0.69	0.51	0.68	1.00	-0.30	-0.33	0.19	-0.07	-0.01	-0.29
O3	-0.14	-0.11	-0.16	-0.45	-0.30	1.00	0.58	-0.44	0.31	0.02	0.29
TEMP	-0.13	-0.10	-0.32	-0.27	-0.33	0.58	1.00	-0.81	0.82	0.04	0.03
PRES	0.02	-0.02	0.22	0.17	0.19	-0.44	-0.81	1.00	-0.75	-0.06	0.07
DEWP	0.11	0.07	-0.26	-0.03	-0.07	0.31	0.82	-0.75	1.00	0.09	-0.30
RAIN	-0.01	-0.03	-0.04	-0.04	-0.01	0.02	0.04	-0.06	0.09	1.00	0.02
WSPM	-0.27	-0.18	-0.11	-0.39	-0.29	0.29	0.03	0.07	-0.30	0.02	1.00

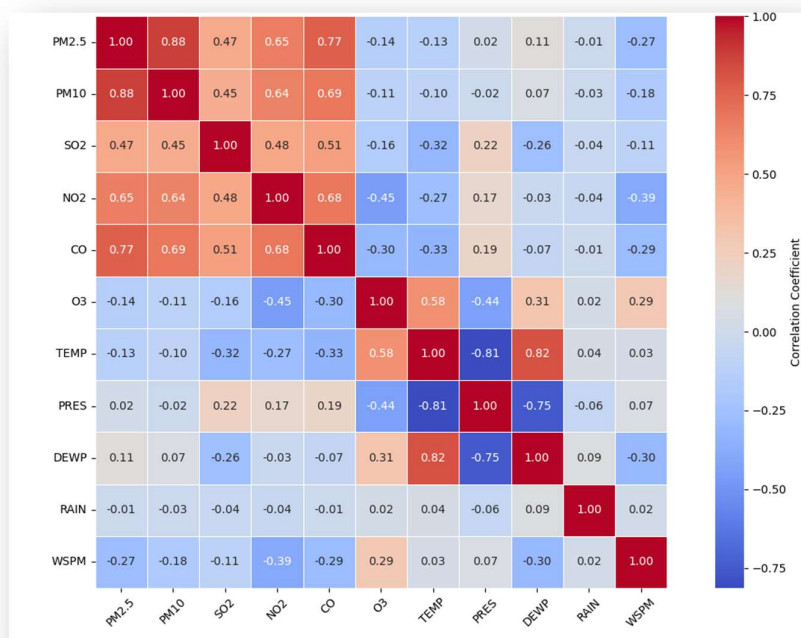


Figure C2

Correlation Heatmap of Air Pollutants and Meteorological Variables

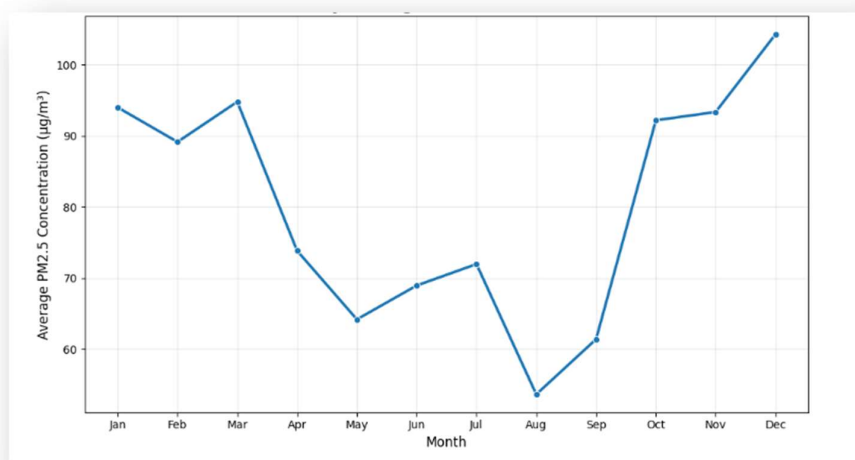


Figure C3

Monthly Average PM2.5 Concentration

Table C2*Average PM2.5 Concentration by Monitoring Station*

	station	PM2.5
0	Dongsi	86.310293
1	Wanshouxigong	85.356562
2	Nongzhanguan	85.289471
3	Gucheng	84.389145
4	Wanliu	83.516054
5	Aotizhongxin	83.163652
6	Guanyuan	83.047023
7	Tiantan	82.255892
8	Shunyi	80.021067
9	Changping	71.119444
10	Huairou	69.949581
11	Dingling	67.433389

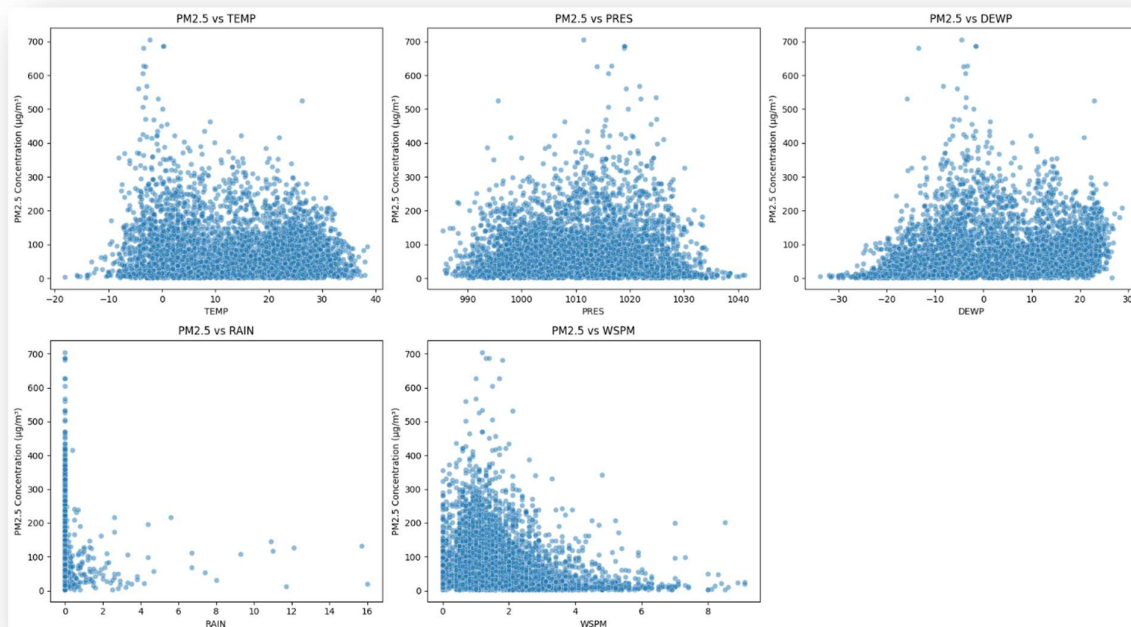
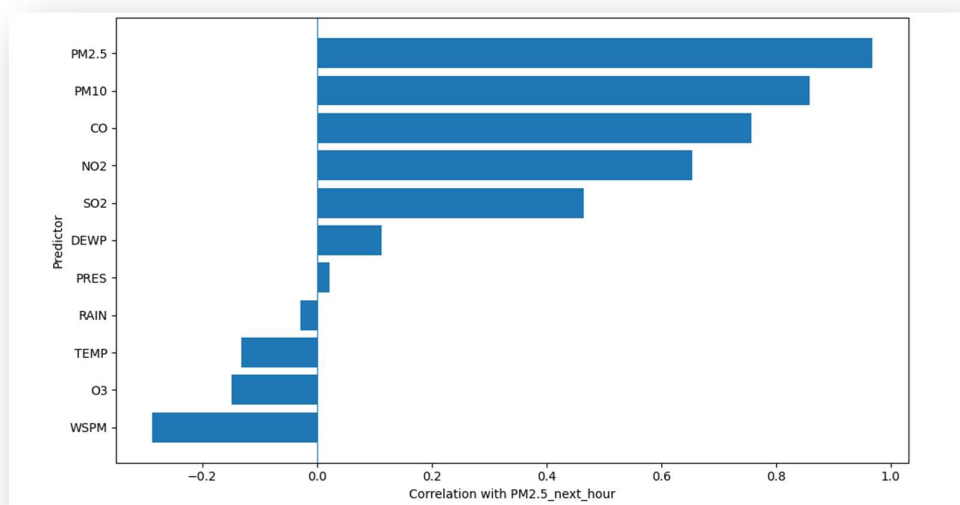
**Figure C4***Relationship Between Meteorological Variables and PM2.5 Concentration*

Table C3

Correlation of Current-Hour Predictors with One-Hour-Ahead PM2.5 Concentration

PM2.5_next_hour	
PM2.5	0.968270
PM10	0.859196
CO	0.757679
NO2	0.654296
SO2	0.464737
DEWP	0.113029
PRES	0.021473
RAIN	-0.028491
TEMP	-0.131560
O3	-0.148029
WSPM	-0.287046

dtype: float64

**Figure C5**

Correlation of Current-Hour Predictors with One-Hour-Ahead PM2.5 Concentration

Forecasting-Specific Exploratory Analysis

To examine the relationship between the current-hour PM2.5 concentration and the forecasting target, the Pearson correlation between current-hour PM2.5 and one-hour-ahead PM2.5 was calculated.

Correlation between current-hour PM2.5 and one-hour-ahead PM2.5: 0.9683

The strong positive correlation indicates substantial temporal continuity in PM2.5 concentrations, supporting the inclusion of current-hour PM2.5 as a predictor for one-hour-ahead forecasting.

Appendix D

Model Development and Predictor Features

Table D1

Predictor Variables Used for One-Hour-Ahead PM_{2.5} Forecasting

Predictor Group	Variables
Air pollutants	PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO, O ₃
Meteorological variables	TEMP, PRES, DEWP, RAIN, WSPM
Temporal variables	month, hour
Wind direction	Dummy-encoded wind-direction categories
Monitoring station	Dummy-encoded monitoring-station categories

Note. The target variable was PM_{2.5}_next_hour, representing PM_{2.5} concentration one hour ahead. Current-hour PM_{2.5} was retained as a predictor because it represents information available at the time the forecast is generated. Following categorical encoding, the final predictor dataset contained 41 features.

Table D2*Machine Learning Models Evaluated*

Model	Model Type	Purpose
Linear Regression	Linear regression	Provide an Interpretable baseline
Decision Tree Regressor	Tree-based	Capture nonlinear relationships and interactions
Random Forest Regressor	Bagging ensemble	Reduce instability and overfitting associated with individual trees
XGBoost Regressor	Gradient boosting	Provide efficient and scalable nonlinear modelling
Gradient Boosting Regressor	Gradient boosting	Improve predictions through sequential error correction
AdaBoost Regressor	Adaptive boosting	Emphasize observations that are more difficult to predict
LightGBM Regressor	Gradient boosting	Provide computationally efficient modelling for large structured data
CatBoost Regressor	Gradient boosting	Broaden comparison of advanced boosting approaches

```

Training set shape: (336578, 41)
Testing set shape: (84156, 41)

Training period:
2013-03-01 00:00:00 to 2016-05-12 17:00:00

Testing period:
2016-05-12 18:00:00 to 2017-02-28 22:00:00

Forecasting horizon: 1 hour ahead
Cutoff time: 2016-05-12 18:00:00

```

Figure D1*Chronological Training and Testing Dataset Split*

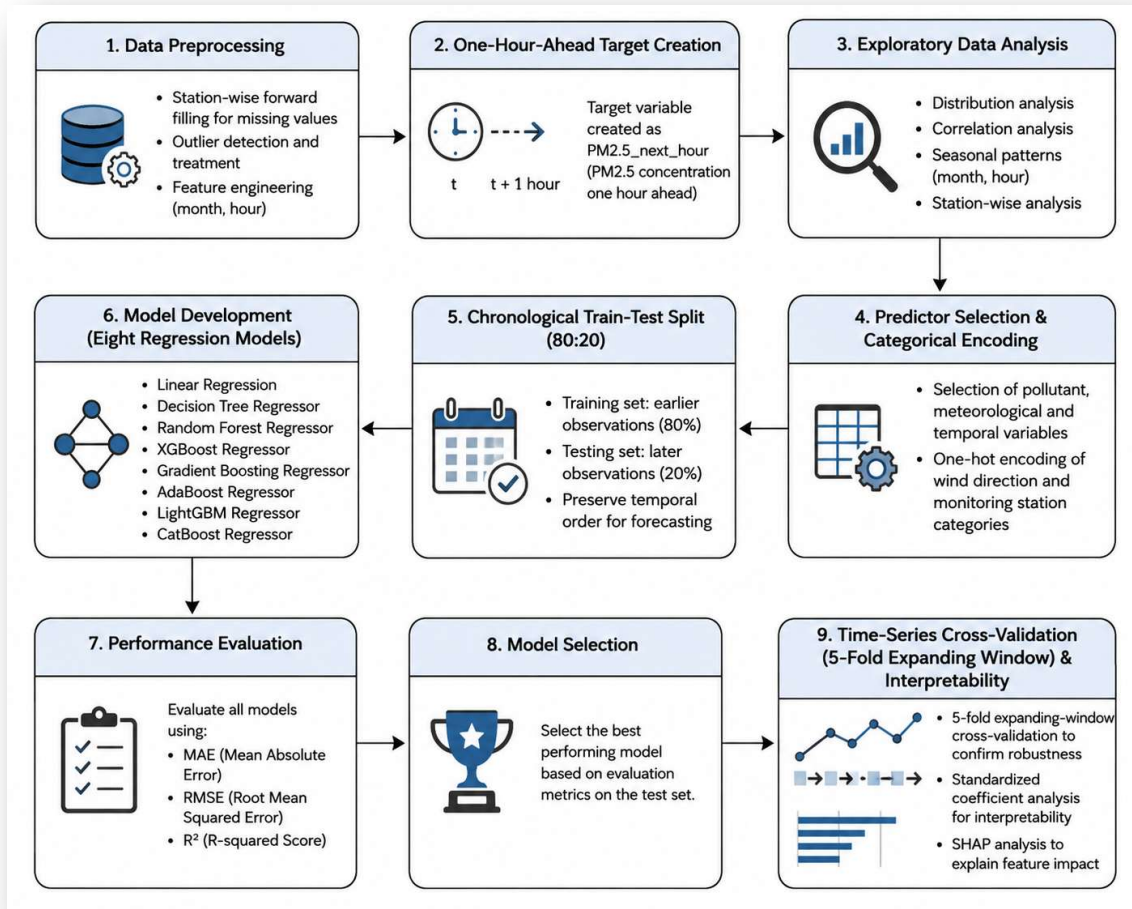


Figure D2

Machine Learning Model Development Workflow

Appendix E

Model Evaluation and Preliminary Results

Table E1

Performance Comparison of the Eight Regression Models

	Model	Train MAE	Train RMSE	Train R ²	Test MAE	Test RMSE	Test R ²
0	Linear Regression	10.81	20.06	0.9382	10.16	19.13	0.9477
1	Gradient Boosting	10.31	19.07	0.9441	9.91	19.48	0.9458
2	LightGBM	9.87	17.79	0.9514	9.85	19.60	0.9452
3	XGBoost	9.79	17.50	0.9530	9.96	19.91	0.9434
4	CatBoost	10.43	19.29	0.9429	10.16	19.94	0.9432
5	Random Forest	3.72	7.03	0.9924	10.40	20.61	0.9394
6	AdaBoost	23.06	30.70	0.8552	22.77	30.58	0.8665
7	Decision Tree	0.00	0.00	1.0000	16.88	31.78	0.8558

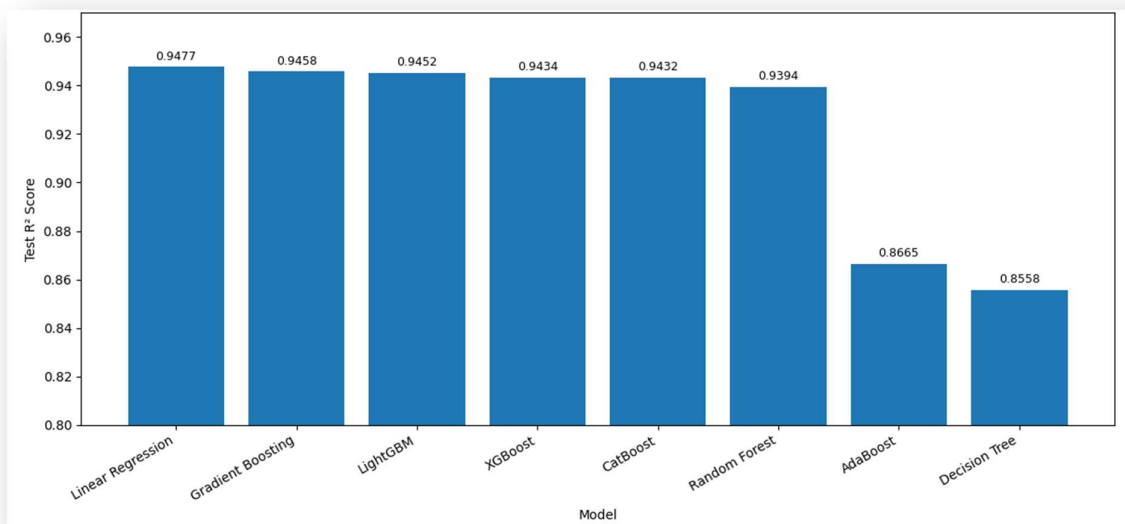


Figure E1

Comparison of Test R² Scores Across Regression Models

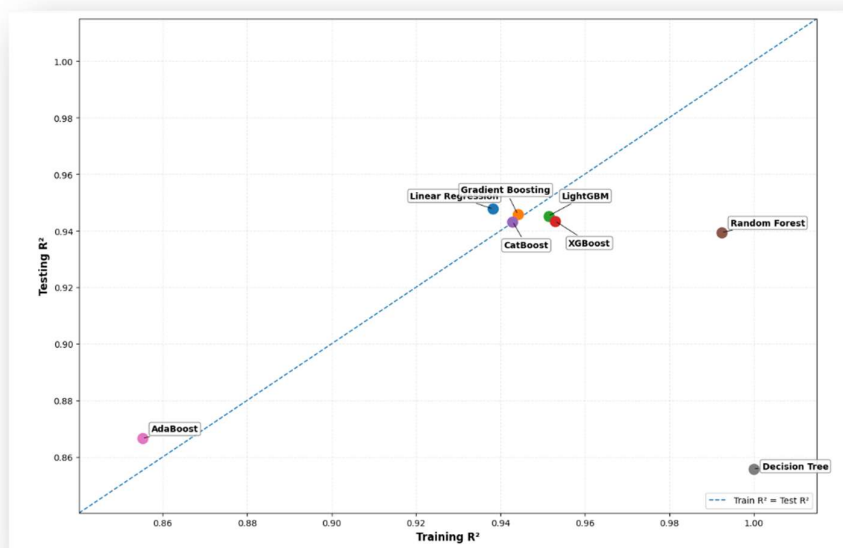


Figure E2

Comparison of Training and Testing R^2 Across Regression Models

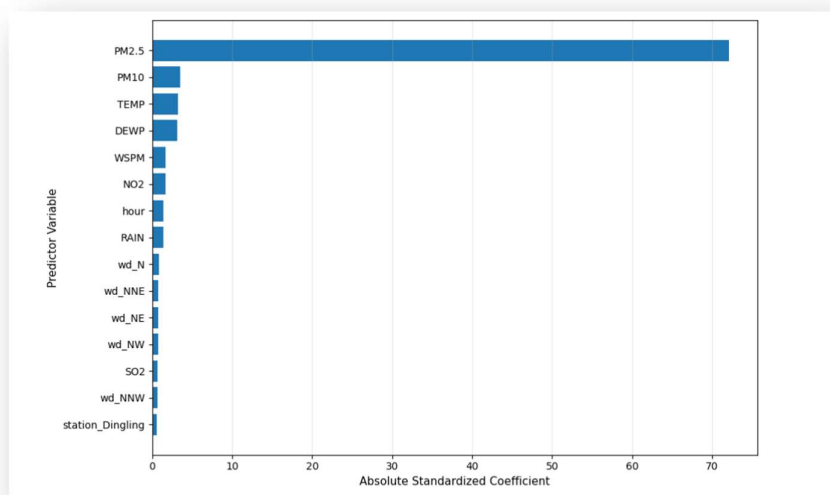


Figure E3

Top 15 Predictors for One-Hour-Ahead PM2.5 Forecasting Based on Absolute Standardized Coefficients

Table E2

Five-Fold Expanding-Window Time-Series Cross-Validation Results for Linear Regression

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LINEAR REGRESSION TIME-SERIES CROSS-VALIDATION
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Fold 1				
	Training observations	:	56,078	
	Validation observations	:	56,100	
	Training period	:	2013-03-01 00:00:00 to 2013-09-11 18:00:00	
	Validation period	:	2013-09-11 19:00:00 to 2014-03-25 13:00:00	
	MAE	:	17.61	
	RMSE	:	25.75	
	R ²	:	0.9258	
Fold 2				
	Training observations	:	112,178	
	Validation observations	:	56,100	
	Training period	:	2013-03-01 00:00:00 to 2014-03-25 13:00:00	
	Validation period	:	2014-03-25 14:00:00 to 2014-10-06 08:00:00	
	MAE	:	10.06	
	RMSE	:	17.33	
	R ²	:	0.9126	
Fold 3				
	Training observations	:	168,278	
	Validation observations	:	56,100	
	Training period	:	2013-03-01 00:00:00 to 2014-10-06 08:00:00	
	Validation period	:	2014-10-06 09:00:00 to 2015-04-19 03:00:00	
	MAE	:	11.60	
	RMSE	:	21.17	
	R ²	:	0.9434	
Fold 4				
	Training observations	:	224,378	
	Validation observations	:	56,100	
	Training period	:	2013-03-01 00:00:00 to 2015-04-19 03:00:00	
	Validation period	:	2015-04-19 04:00:00 to 2015-10-30 22:00:00	
	MAE	:	8.56	
	RMSE	:	14.54	
	R ²	:	0.9314	
Fold 5				
	Training observations	:	280,478	
	Validation observations	:	56,100	
	Training period	:	2013-03-01 00:00:00 to 2015-10-30 22:00:00	
	Validation period	:	2015-10-30 23:00:00 to 2016-05-12 17:00:00	
	MAE	:	11.04	
	RMSE	:	23.38	
	R ²	:	0.9468	
=====				
CROSS-VALIDATION SUMMARY				
=====				
	Fold	MAE	RMSE	R ²
0	1	17.61	25.75	0.9258
1	2	10.06	17.33	0.9126

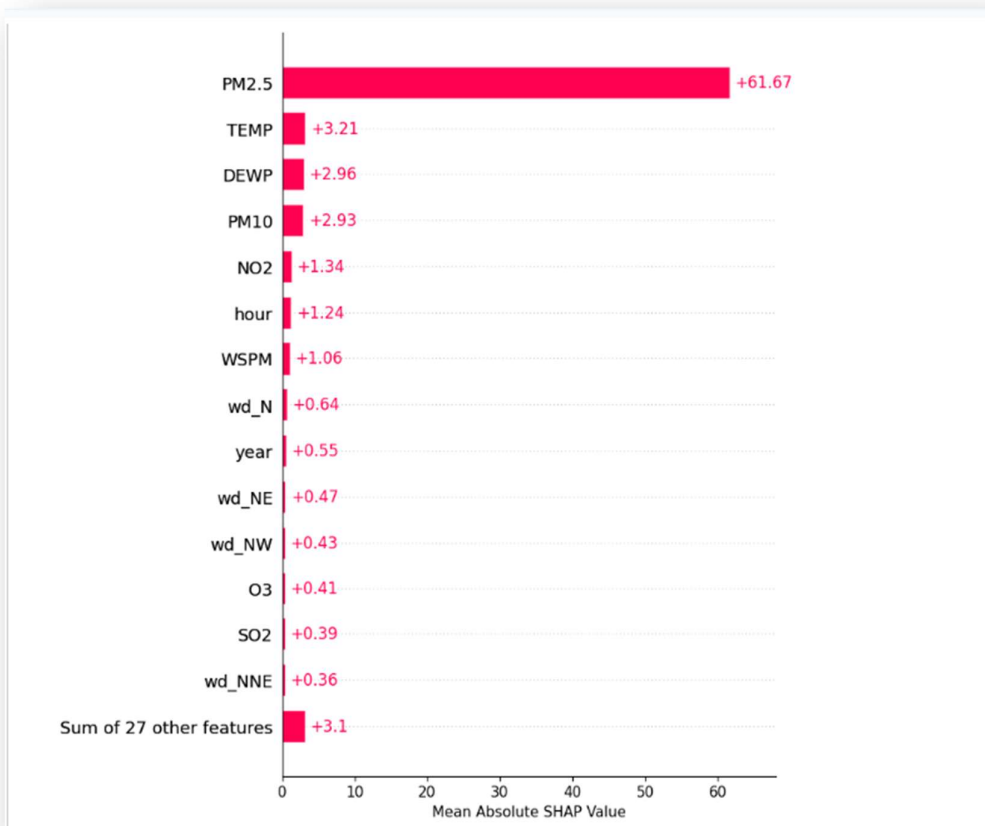


Figure E4

SHAP Feature Importance for One-Hour-Ahead PM2.5 Forecasting

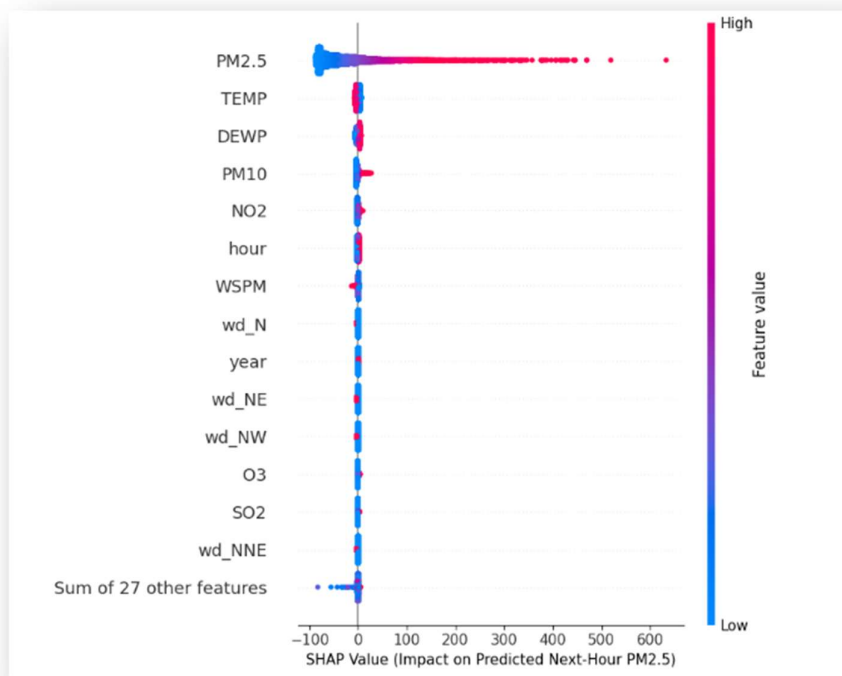


Figure E5

SHAP Summary Plot for One-Hour-Ahead PM2.5 Forecasting

Appendix F

Minimum Sample Size Computation by Research Question

Table F1

Minimum Sample Size Requirement by Research Question

Research Question	Minimum Required Sample Size	Available Aligned Observations	Adequacy
RQ1	385	420,734	Adequate
RQ2	385	420,734	Adequate
RQ3	385	420,734	Adequate
RQ4	385	420,734	Adequate

Note. The minimum sample size of 385 observations was calculated using a 95% confidence level ($Z = 1.96$), a 5% margin of error ($e = .05$), and a conservative estimated population proportion ($p = .50$). Following preprocessing and construction of the one-hour-ahead PM2.5 forecasting target, 420,734 aligned observations were available for analysis. The available sample therefore substantially exceeded the minimum requirement for all four research questions.